Better Research Brighter Future

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LEARN PRACTICE EXCEL

RESEARCH METHODOLOGY & BIOSTATISTICS

THE COMPLETE STUDY GUIDE

HIGH-YIELD TOPICS • MBBS | BHMS | B.Sc. NURSING | BDS | BPT

Easy • Quick • Exam-Oriented Notes

Data into Knowledge



KEY AREAS

- Research Process
- Research Design
- Sampling Techniques
- Data Collection Methods
- Questionnaire & Tools
- Validity & Reliability
- Data Analysis
- Parametric & Non-Parametric Tests
- Correlation & Regression
- Chi-Square Test
- Interpretation & Report Writing

WHY THIS BOOK?

- High-Yield Points
- Concept Clarity
- Diagrams & Flowcharts
- Previous Year Exam Focus
- Easy to Revise

Research Today Better Care Tomorrow

RESEARCH PROCESS | STUDY DESIGN | SAMPLING TECHNIQUES | DATA ANALYSIS | INFERENTIAL STATISTICS | REPORT WRITING

For Medical Students & Future Healthcare Professionals

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Study Smart Grow

RESEARCH METHODOLOGY & BIOSTATISTICS

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Research Methodology & Biostatistics

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 49

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

INDEX

1. INTRODUCTION TO RESEARCH
   - 1.1 Definition of Research
   - 1.2 History of Research
   - 1.3 Objectives of Research
   - 1.4 Scope of Research
   - 1.5 Research Methods
   - 1.6 Research Methodology
   - 1.7 Research Method vs Research Methodology
   - 1.8 Criteria of Good Research
2. RESEARCH PROBLEM
   - 2.1 Definition / Meaning
   - 2.2 Statement of Research Problem
   - 2.3 Purpose of Research Problem
   - 2.4 Objectives of Research Problem
   - 2.5 Identification of Research Problem
   - 2.6 Characteristics of Good Research Problem
3. RESEARCH DESIGN
   - 3.1 Meaning / Definition
   - 3.2 Need / Importance
   - 3.3 Features of Good Research Design
   - 3.4 Types of Research Design
   - 3.5 Basic Principles of Research Design
4. INFORMED CONSENT
   - 4.1 Definition
   - 4.2 Purpose and Importance
   - 4.3 Essential Elements
   - 4.4 Rights of Participants
5. RESEARCH PROPOSAL
   - 5.1 Definition
   - 5.2 Purpose
   - 5.3 Components / Structure
   - 5.4 Importance
   - 5.5 Preparation of Research Proposal
6. RESEARCH FUNDING
   - 6.1 Meaning
   - 6.2 Sources of Funding
   - 6.3 Significance
7. BIAS IN RESEARCH
   - 7.1 Definition
   - 7.2 Types of Bias
   - 7.3 Sources / Causes
   - 7.4 Effects on Research
   - 7.5 Methods to Minimize Bias
8. ETHICAL PRINCIPLES IN RESEARCH
   - 8.1 Meaning of Research Ethics
   - 8.2 Respect for Participants
   - 8.3 Confidentiality and Privacy
   - 8.4 Beneficence and Non-maleficence
   - 8.5 Justice
   - 8.6 Voluntary Participation
   - 8.7 Ethical Approval / Ethics Committee
9. SAMPLING DESIGN
   - 9.1 Definition / Meaning
   - 9.2 Criteria for Selecting Sampling Procedure
   - 9.3 Steps in Sampling Design
   - 9.4 Characteristics of Good Sample Design
10. TYPES OF SAMPLING
    - 10.1 Probability Sampling
      - 10.1.1 Simple Random Sampling
      - 10.1.2 Systematic Sampling
      - 10.1.3 Stratified Sampling
      - 10.1.4 Cluster Sampling
      - 10.1.5 Multistage Sampling
    - 10.2 Non-probability Sampling
      - 10.2.1 Convenience Sampling
      - 10.2.2 Purposive / Judgment Sampling
      - 10.2.3 Quota Sampling
      - 10.2.4 Snowball Sampling
    - 10.3 Probability vs Non-probability Sampling
11. MEASUREMENT
    - 11.1 Meaning / Definition
    - 11.2 Importance of Measurement
    - 11.3 Measurement Scale
    - 11.4 Sources of Error in Measurement
    - 11.5 Reliability and Validity of Measurement
12. MEASUREMENT SCALES
    - 12.1 Nominal Scale
    - 12.2 Ordinal Scale
    - 12.3 Interval Scale
    - 12.4 Ratio Scale
    - 12.5 Characteristics and Examples
    - 12.6 Comparison of Scales
13. DEVELOPMENT OF MEASUREMENT TOOL
    - 13.1 Meaning of Measurement Tool
    - 13.2 Steps of Tool Development
    - 13.3 Selection of Items
    - 13.4 Reliability
    - 13.5 Validity
    - 13.6 Standardization
14. SCALING
    - 14.1 Meaning / Definition
    - 14.2 Importance
    - 14.3 Types of Scaling
    - 14.4 Characteristics of Good Scale
15. METHODS OF DATA COLLECTION
    - 15.1 Primary Data
    - 15.2 Secondary Data
    - 15.3 Schedule
    - 15.4 Questionnaire
    - 15.5 Interview
16. SCHEDULE
    - 16.1 Definition
    - 16.2 Purpose
    - 16.3 Essentials of Good Schedule
    - 16.4 Advantages
    - 16.5 Limitations
17. QUESTIONNAIRE
    - 17.1 Definition
    - 17.2 Types
    - 17.3 Construction / Preparation
    - 17.4 Response / Non-response Problems
    - 17.5 Reliability
    - 17.6 Validity
    - 17.7 Advantages
    - 17.8 Limitations
    - 17.9 Questionnaire vs Schedule
18. REVIEW OF LITERATURE
    - 18.1 Definition
    - 18.2 Purpose and Importance
    - 18.3 Sources of Literature
    - 18.4 Steps of Literature Review
    - 18.5 Identification of Research Gap
19. INTERVIEW
    - 19.1 Definition
    - 19.2 Purpose
    - 19.3 Types of Interview
    - 19.4 Interview Procedure
    - 19.5 Advantages
    - 19.6 Limitations
    - 19.7 Interviewer Bias
20. RELIABILITY
    - 20.1 Definition
    - 20.2 Importance
    - 20.3 Types of Reliability
      - 20.3.1 Test-retest Reliability
      - 20.3.2 Inter-rater Reliability
      - 20.3.3 Intra-rater Reliability
      - 20.3.4 Internal Consistency
    - 20.4 Factors Affecting Reliability

BPT23

BIOSTATISTICS

Page - 1

:: INDEX ::

UNIT - I : Statistics & Presentation of Data :

1. Statistics :
   - Definition, Characteristics, Importance
   - Branches (Descriptive & Inferential)
   - Applications in Physiotherapy
2. Presentation of Data :
   - Meaning, Objectives, Basic Principles
   - Types (Textual, Tabular, Diagrammatic, Graphical)
3. Variables and Their Types :
   - Definition, Types, Qualitative, Quantitative
   - Discrete, Continuous, Independent, Dependent
4. Measurement Scales :
   - Nominal, Ordinal, Interval, Ratio
   - Characteristics, Examples, Comparison
5. Types of Diagrams :
   - Bar Diagram, Pie Diagram, Pictogram
   - Other diagrams
6. Graphs :
   - Principles, Techniques of construction
   - Histogram, Frequency Polygon, Frequency Distribution Graph
   - Interpretation
7. Limitations of Diagrams and Graphs :
   - Limitations of diagrams, Limitations of graphs
   - Common errors

UNIT - II : Central Tendency, Probability & Distribution :

8. Central Tendency :
   - 8.1 Meaning & Need :
     - Definition, Meaning, Importance
   - 8.2 Mean :
     - Types, Arithmetic Mean, Formula
     - Calculation, Advantages, Limitations
   - 8.3 Median :
     - Definition, Calculation, Advantages, Limitations, Uses
   - 8.4 Mode :
     - Definition, Calculation, Advantages, Limitations, Uses
   - 8.5 Comparison :
     - Mean vs Median vs Mode, When to use
9. Probability :
   - Meaning, Basic concept, Types / Rules
   - Applications
10. Standard Distributions :
    - 10.1 Normal Distribution :
      - Meaning, Characteristics, Properties
      - Normal curve, Applications
    - 10.2 Divergence from Normality :
      - Meaning, Causes, Interpretation
      - Skewness, Kurtosis
    - 10.3 Skewness :
      - Meaning, Symmetrical, Positive, Negative
    - 10.4 Kurtosis :
      - Meaning, Types, Interpretation

UNIT - III : Correlation, Tests & SPSS :

11. Correlation Analysis :
    - Meaning, Purpose, Types (Positive, Negative, Zero, Linear / Non-linear)
12. Methods of Correlation :
    - 12.1 Karl Pearson's Coefficient :
      - Meaning, Formula, Calculation
      - Interpretation, Uses, Limitations
    - 12.2 Rank Correlation Coefficient :
      - Spearman's Rank Correlation
      - Formula, Calculation, Interpretation
      - Uses, Limitations
13. Scatter Diagram :
    - Definition, Purpose, Construction
    - Interpretation, Types / Patterns
14. Parametric Tests :
    - Meaning, Characteristics, Assumptions
    - Common Tests, Selection / Application
    - Advantages, Limitations
15. SPSS :
    - Introduction, Uses, Data entry
    - Variable definition, Data analysis
    - Descriptive statistics, Interpretation

UNIT - IV : Sampling, Significance & ANOVA :

16. Sampling Techniques :
    - Definition, Need, Criteria, Procedure, Design
    - Types (Probability & Non-probability)
    - Simple random, Systematic, Stratified, Cluster
    - Convenience, Purposive etc.
17. Sampling Errors :
    - Meaning, Causes, Types / Sources
    - Methods to reduce
18. Sampling Variation :
    - Meaning, Causes, Sampling distribution
    - Importance
19. Test of Significance :
    - Meaning, Purpose, Null & Alternative Hypothesis
    - Level of Significance, p-value, Critical value
    - Statistical significance, Interpretation
    - Type I & Type II errors
20. Analysis of Variance (ANOVA) :
    - Meaning, Purpose, Basic principle
    - Within & between groups, F-ratio
    - Interpretation, Applications, Assumptions
    - Limitations, Types (One-way, Two-way)
21. Analysis of Covariance (ANCOVA) :
    - Meaning, Purpose, Basic principle, Covariate
    - Difference between ANOVA & ANCOVA
    - Applications, Interpretation

5. CENTRAL TENDENCY

5.1 Central Tendency :-

Definition :- Central tendency refers to a single value that represents the entire set of data. It is the average or typical value around which the data tends to cluster.

Meaning :-
It gives a central value of the data and helps to summarize large data in a single figure.

Importance :-
- Helps in understanding the data in a simple form.
- Useful for comparison between groups.
- Widely used in research, clinical practice and
- decision making.
- Helps in further statistical analysis.

5.2 Mean :-

Definition :- Mean is the sum of all the values divided by the number of observations. It is also called arithmetic mean.

Formula :- $\bar{X} = \frac{\Sigma X}{N}$

ΣX = Sum of all observations
N = Number of observations

Types of Mean :-
1. Simple / ungrouped data
2. Frequency distributions
3. Discrete frequency table
4. Grouped / continuous data
5. Missing value.

1. Simple / Ungrouped data :-

Question :- Find the mean of the following data.
4, 7, 9, 12, 18

Solution :-

$\Sigma X = 4 + 7 + 9 + 12 + 18 = 50$
$N = 5$

$\text{Mean } (\bar{X}) = \frac{\Sigma X}{N} = \frac{50}{5} = 10$

$\therefore$ Mean = 10

2. Frequency distribution :-

Question :- Find the mean of the following data.

Marks (x)	10	20	30	40	50
Frequency (f)	2	5	8	10	5

Solution :-

$\Sigma fx = (10\times2) + (20\times5) + (30\times8) + (40\times10) + (50\times5)$
$= 20 + 100 + 240 + 400 + 250 = 1010$

$\Sigma f = 2 + 5 + 8 + 10 + 5 = 30$

$\text{Mean } (\bar{X}) = \frac{\Sigma fx}{\Sigma f} = \frac{1010}{30} = 33.67$

$\therefore$ Mean = 33.67

3. Discrete frequency table :-

Question :- Find the mean of the following data.

Value (x)	1	2	3	4	5
Frequency (f)	4	7	6	2	1

Solution :-

$\Sigma fx = (1\times4) + (2\times7) + (3\times6) + (4\times2) + (5\times1)$
$= 4 + 14 + 18 + 8 + 5 = 49$

$\Sigma f = 4 + 7 + 6 + 2 + 1 = 20$

$\text{Mean } (\bar{X}) = \frac{\Sigma fx}{\Sigma f} = \frac{49}{20} = 2.45$

$\therefore$ Mean = 2.45

4. Grouped/ Continuous data :-

Question :- Find the mean of the following data.

Class interval	Frequency (f)	Mid value (x)	fx
0 – 10	5	5	25
10 – 20	8	15	120
20 – 30	12	25	300
30 – 40	10	35	350
40 – 50	5	45	225
Total	40		1020

Solution :-

$\Sigma fx = 1020 \quad \Sigma f = 40$

$\text{Mean } (\bar{X}) = \frac{\Sigma fx}{\Sigma f} = \frac{1020}{40} = 25.5$

$\therefore$ Mean = 25.5

5. Missing value :-

Question :- The mean of 5 numbers is 20. If the four numbers are , 18, 22, 24 and 26, find the missing number.

Solution :-

$\text{Mean } (\bar{X}) = \frac{\Sigma X}{N} \Rightarrow 20 = \frac{12 + 18 + 24 + 26 + x}{5}$

$100 = 80 + x \Rightarrow x = 100 - 80 = 20$

$\therefore$ Missing number = 20

5.3 Median :-

Definition :- Median is the middle value when the data is arranged in ascending or descending order.

Types :-
1. Odd number of observations
2. Even number of observations
3. Frequency distribution
4. Grouped / continuous data

Formula :-

For odd N : $\text{Median} = \frac{N+1}{2}$ th term

For even N : $\text{Median} = \frac{\left(\frac{N}{2}\right)\text{th term} + \left(\frac{N}{2}+1\right)\text{th term}}{2}$

THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

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